

Part 2 of 2

Plasma Arc Gasification For Waste Management

A plasma pyrolysis system developed in Gujarat can treat 15-20 kg of hospital waste per hour. It can be set up at a cost of Rs 1-1.2 million. The method can be used efficiently for municipal solid waste and plastic waste also

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Part 1 of this article dealt with plasma gasification, treatment of wastes in plasma gasifiers and components of plasma gasification systems. In continuation, I describe here some of the plasma gasification systems used worldwide and in India.

Plasma gasification systems used worldwide

Some of the well-known global firms are Westinghouse Plasma Corporation (WPC), Integrated Environmental Technologies (IET), The Solena Group and Startech. Their proprietary plasma gasification systems are described below.

Westinghouse Plasma Corporation (WPC). WPC's proprietary plasma gasification vitrification reactor design (Fig. 5) combines a moving-bed gasifier with its industrial plasma torch technology. WPC, in conjunction with Alter Nrg (an engineering firm), is able to provide full gasification equipment, complete with feed handling, slag removal and gas cooling systems, for various applications. The system engineering drawing is shown in Fig. 6.

WPC has commercially operating/under-installation facilities in Japan, USA and India.

Japan. Hitachi Metals's waste-to-

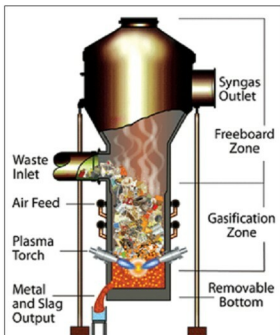


Fig. 5: Westinghouse Plasma Corporation's proprietary plasma gasification vitrification reactor

energy facility, located in Utashinai, Japan, was constructed in 2002 and has been operational since 2003. The facility processes 200-280 tonnes/day of a mixture of auto shredder residue and municipal solid waste to produce electricity.

A second facility, built near the neighbouring cities of Mihama and Mikata, Japan, was commissioned in 2002 to treat 20 tonnes per day of municipal solid waste and four tonnes per day of sewage sludge for production of heat utilised in a municipal waste-water

treatment plant.

India. Two hazardous waste disposal plants with a capacity of 72 tonnes/day and using WPC gasification technology and plasma torches to produce electricity are under construction in Pune and Nagpur, by SMS Infrastructures, India. When operational, these are expected to produce 5 megawatts of electricity.

USA. One WTE facility is at approval stage in the US. When completed, Geoplasma's WTE plant in Florida will be the largest plasma gasification facility in the world processing up to 3000 tonnes per day of municipal solid waste and producing 120 MW of electricity. The first phase will process 1500 tonnes per day and produce 60 MW of electricity, which is enough to power 60,000 homes. The facility is expected to become operational by mid-2010.

Integrated Environmental Technologies (IET). It uses a plasma-enhanced melter (PEM) to transform waste into highly stable glass-like residues, recoverable metals and hydrogen-enriched gas (Fig. 7). The technology is unique in that it combines three processes: plasma arc